

Prem Patel

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EDUCATION

University of Waterloo

Bachelor of Computer Science | GPA: 3.7

Waterloo, ON

Expected Apr. 2027

Wilfrid Laurier University

Bachelor of Business Administration | President's Gold Scholarship

Waterloo, ON

Expected Apr. 2027

EXPERIENCE

Cloudflare

Software Engineer Intern, Distributed Systems

Incoming Fall 2026

Austin, TX

Tesla

Software Engineer Intern, Distributed Systems

May 2026 – Aug. 2026

Palo Alto, CA

- Built a **scalable analytics backend** for Powerwall residential batteries enrolled in **Virtual Power Plant** programs (**Rust, Scala**), reducing response time for VPP partner escalations by **90%**
- Iterated on an **ETL prototype** with user feedback, then **re-architected** and built a **production system** with a **gRPC API**, indexed **PostgreSQL** schema, **Spark**-backed data-writing layer, and client auth/access controls
- Developed distributed **Scala services** to aggregate and process performance data from **millions of Powerwall IoT devices**, deploying through **Argo CD** and **Kubernetes**
- Finalized the team's object-storage migration and deployed changes to production on my **first day**

Kinaxis

Software Engineer Intern, Embedded Algorithms

Jan. 2025 – Apr. 2026

Ottawa, ON

- Built end-to-end features in a **distributed C++ server** with **TypeScript (Node.js)** + **Java** agents/workers over persistent **WebSockets**, improving reliability, latency, and throughput under load
- Shipped **BYO-Model** with a solver-agnostic, **async job pipeline** to run optimization models at scale
- Designed thread-safe CPU/memory **guardrails** in **C++** using **RAII semaphores** to prevent noisy-neighbor resource spikes in **multi-tenant** clusters; cut lock contention by **32%**
- Implemented an **IPC state handler** with retry scheduling and **disconnection detection**, improving **fault tolerance** and preventing cascading failure logs after worker/agent disconnects
- Improved response serialization with **FlatBuffers** to support **25%** larger payloads and eliminated silent **Node.js** crashes, increasing throughput and reliability under load
- Developed a **VS Code extension** to author, publish, and run/debug user-defined **embedded algorithms (EA)** with breakpoints, reducing developer iteration time by **40%**
- Standardized **structured logging** + metrics integrated with Datadog dashboards to attribute CPU usage per customer, enabling transparent **consumption-based pricing** generating **\$1M+** annual revenue
- Deployed EA services on **Kubernetes** via **Docker** + **Helm (Azure/GCP)** and presented **17** cross-team demos

AutoTrader

Software Engineer Intern, Data Automation

Jan. 2024 – May 2024

Toronto, ON

- Developed **Python** ingestion services processing OEM feeds for **30+ automotive brands**; reduced end-to-end processing time by **8x** by restructuring transformations and batching (Pandas + REST API integrations)
- Built **Azure Pipelines** automation to generate and deliver monthly reports for **150+ dealerships**, integrating **Redash** queries and enforcing consistent schemas/validation
- Reduced manual cleanup by **5 hrs** per refresh cycle and improved reporting accuracy by up to **85%**

TECHNICAL SKILLS

Languages: C++, Java, Scala, Python, TypeScript (JavaScript), Rust, SQL, Bash, C, C#

Frameworks/Libs: gRPC, Spring Boot (DI, WebSockets), Apache Spark, React, FastAPI, Boost, FlatBuffers

Dev Tools: Git, Node.js, Docker, Kubernetes, Helm, Argo CD, PostgreSQL, Jenkins, Datadog, GCP, Azure, AWS